

UCS503 - Software Engineering

Estate Desk

A WhatsApp-Based AI Real-Estate Lead and Meeting Assistant

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Project Proposal

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Introduction

1.1 Background and Context

Property buyers already use WhatsApp to ask about price, location, bedrooms, amenities and site visits. A small real-estate team may receive many such conversations at the same time. Important details are spread across long chat threads, so a manager must repeatedly read messages before deciding which buyer needs attention. This delays replies and can cause a useful lead to be missed.

Estate Desk is proposed as a WhatsApp-first customer engagement platform. The buyer continues to use WhatsApp, while the property manager uses a web dashboard. A narrowly scoped OpenClaw agent answers only real-estate questions, collects buyer requirements, recommends properties from an uploaded catalog and asks interested buyers whether they want a physical property visit. The dashboard turns the permitted chat into simple lead, meeting and trend views.

1.1.1 Problem Statement

The project addresses four connected problems:

- buyer needs such as budget, preferred sector, bedroom count and buying timeline are written in different ways across many messages;
- managers may not know which available properties best match a buyer;
- interest in a physical visit may not be noticed at the right time; and
- normal chatbots may answer unrelated questions, invent property facts or mix the context of different buyers.

The proposed system must therefore be useful without becoming an uncontrolled general-purpose assistant. It will use only authorized contacts and uploaded property data. Prices, legal facts, availability and final meeting details will always be checked by a human manager.

1.2 Project Objectives

1. **Build a WhatsApp-first property agent:** connect one manager's WhatsApp account to OpenClaw and keep separate context for every buyer.

2. **Collect structured requirements:** identify budget, BHK, area, property type, timeline and visit interest from normal conversation.
3. **Recommend grounded properties:** suggest only records that exist in the manager's uploaded property catalog.
4. **Give managers live guidance:** update overview, leads, meetings and trends pages as valid WhatsApp events arrive.
5. **Protect scope and consent:** refuse unrelated tasks, block unknown contacts, obey STOP and require a human to confirm important actions.

Figure 1.1 shows the complete user journey at a simple level. The website is not a second chat application; it helps the manager understand and act on the WhatsApp conversation.

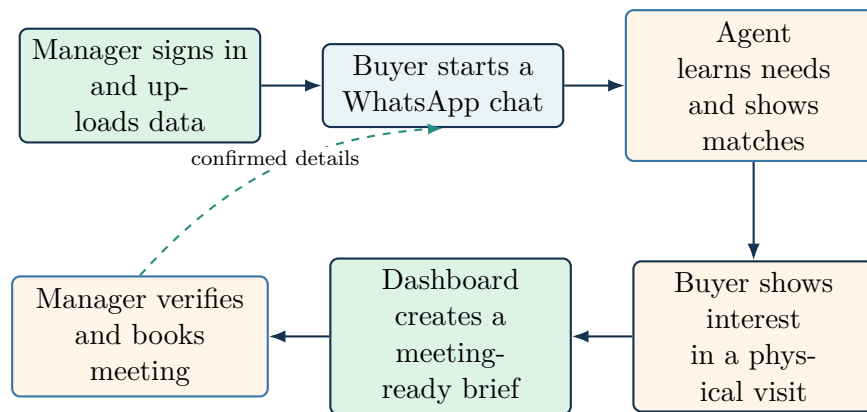


Figure 1.1: Estate Desk user journey from setup to a human-confirmed visit.

1.3 Project Scope

The prototype will support phone-number sign-up and login, CRM and listing CSV uploads, a separate WhatsApp real-estate personality, signed message-event ingestion, lead summaries, property matching, meeting readiness, trend analysis and CSV exports. The supplied historical Gurgaon listings are sample data and must not be presented as live offers.

Direct contracts with commercial CRM vendors, payment collection, legal advice, property ownership verification and automatic meeting confirmation are outside the first version. These limits keep the student project practical and make its safety rules clear.

Methodology and Design

2.1 System Architecture

The system uses two connected parts. OpenClaw runs the WhatsApp channel and the real-estate agent. Estate Desk stores workspace data and shows results to the manager. A signed event hook sends successful incoming and outgoing message events from OpenClaw to the web application. This separation makes it possible to change the model or dashboard without rebuilding the complete system.

2.1.1 High-Level Design

Figure 2.1 uses layers to show where each responsibility belongs. Contact authorization and the real-estate-only policy are applied before an answer is sent. The web application validates every event again before it updates the database or manager screens.

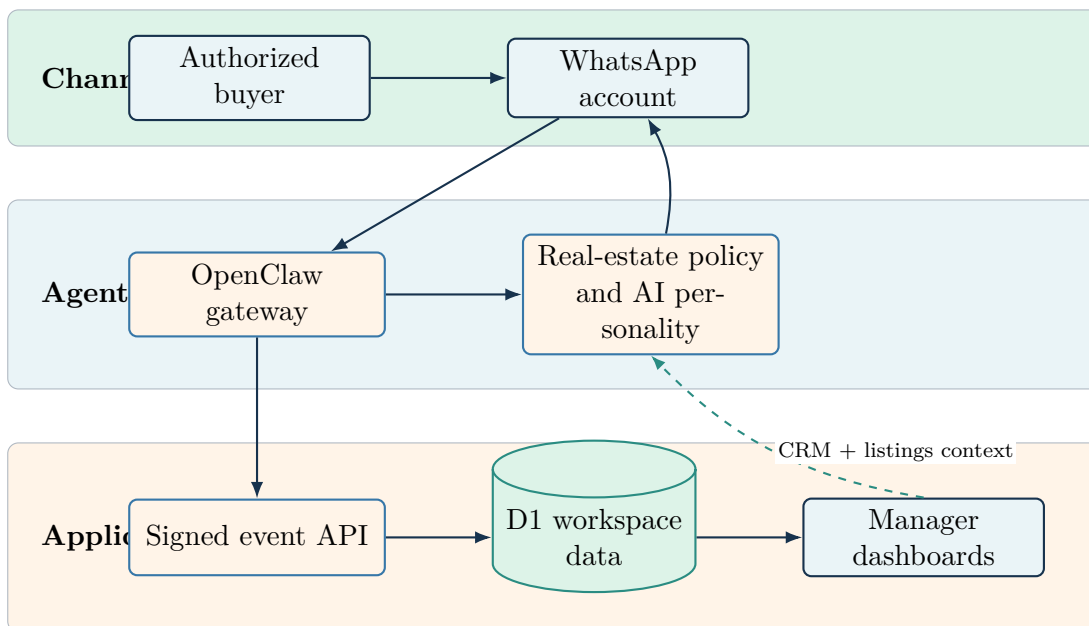


Figure 2.1: Layered architecture for the WhatsApp agent and manager dashboard.

2.1.2 Technical Stack

- **Framework and language:** Vnext/React with TypeScript for the web interface and server routes.
- **Database:** Cloudflare D1 for workspaces, contacts, listings, message events, buyer facts and property matches.
- **AI and messaging:** OpenClaw for WhatsApp routing and a separate real-estate agent identity with per-contact sessions.
- **Live updates:** server-sent events so open dashboards refresh after a valid message is processed.
- **Data exchange:** CSV upload and export for the prototype CRM and property catalog.
- **Testing and version control:** TypeScript checks, domain tests, API smoke tests, production builds and GitHub.

The design uses familiar and mature web tools, so the team can focus on workflow, safety rules and evaluation. Cloudflare D1 provides an SQL model that fits contacts, listings and message events [Clo26]. WhatsApp business messaging requires clear user choice and responsible handling [Wha26].

2.2 Implementation Details

2.2.1 Module 1: Onboarding and Workspace Data

The landing page asks the manager to sign up or log in with a phone number. A new workspace cannot open the full dashboard until the manager supplies two datasets. The CRM CSV contains authorized client phone numbers and consent states. The listing CSV contains property ID, location, price, BHK, area, property type and availability notes. Managers may also add a client or property through a form. Each workspace is kept separate from every other workspace.

The prototype provides downloadable templates and a normalized sample catalog from a historical Gurgaon dataset. Import checks reject missing columns, invalid phone numbers and badly formed price values. Duplicate contacts and listing IDs are updated carefully instead of being copied many times.

2.2.2 Module 2: WhatsApp Event and Insight Pipeline

Each received or successfully sent WhatsApp message gets a unique event ID. The hook includes the account, channel, contact, direction, time and message text in a signed request. The API checks the signature, workspace, contact, channel, event size and ID. Repeated IDs are ignored. This gives the dashboard a clean event ledger and avoids counting one message twice.

After a valid event is stored, the analysis service updates the buyer profile, lead stage and property ranking. Figure 2.2 is a sequence-style flowchart: time moves from left to right, while each vertical line represents a different part of the system.

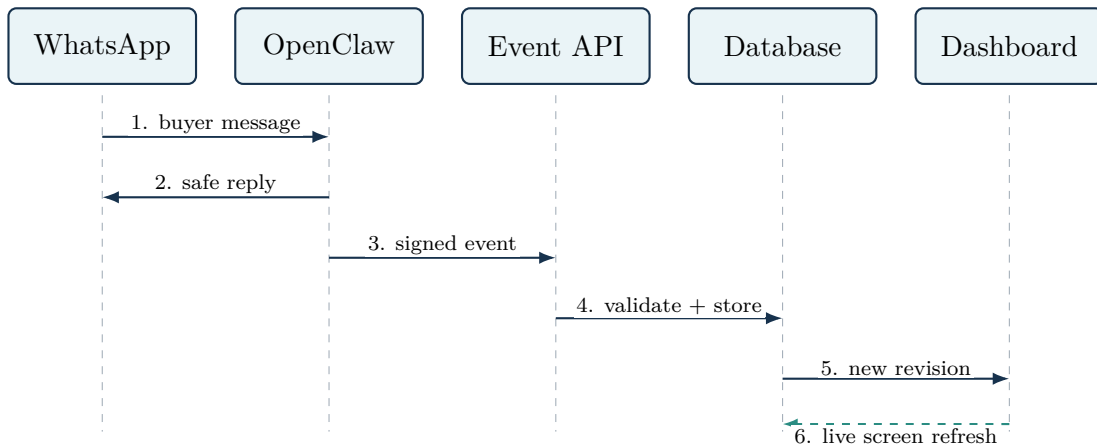


Figure 2.2: Near-real-time message sequence from WhatsApp to the dashboard.

2.2.3 Module 3: Real-Estate Agent and Safety Rules

The agent asks one or two useful questions at a time and keeps a separate conversation for each contact. Its goal is to understand the buyer and, when there is enough interest, ask whether the buyer wants a physical visit at one or more matching properties. It must never invent a listing, promise availability, confirm a booking or discuss unrelated topics.

Safety does not depend only on the personality prompt. A second policy layer checks the channel, recipient, consent state and outgoing content. Unknown contacts are blocked. An inbound-only contact may receive replies but no cold outreach. STOP changes the contact to opted-out immediately. Suspicious requests receive a short real-estate scope reminder. This layered approach follows the general idea that prompt-injection risk should be handled with validation, least privilege and clear boundaries [OWA26].

2.2.4 Module 4: Manager Dashboards

The **Overview** page shows lead counts, conversation activity and stage distribution. The **Leads** page lists buyer details, known requirements, matched properties and the recommended next action. The **Meetings** page shows “buyer ready for physical meeting” cards with the buyer’s contact, requirements, useful chat facts and one or more properties to visit. The **Trends** page compares chat demand with the listing database to show the most requested sectors, BHK values and specific properties. A CSV export combines agent analysis and permitted conversation data for manager use.

Results and Evaluation

3.1 Testing Methodology

This is a project proposal, so the following tests describe how the prototype will be evaluated rather than claiming final production results.

- **Unit Testing:** test phone normalization, CSV validation, requirement extraction, consent changes and property scoring separately.
- **Integration Testing:** send signed sample WhatsApp events and confirm that messages, lead facts, matches and live screens update together.
- **Performance Testing:** compare event timestamps with dashboard revision timestamps and measure the delay under a small pilot load.
- **Safety Testing:** try unrelated questions, prompt-injection text, unknown recipients, duplicate events and STOP/START messages.
- **User Testing:** ask 5–10 consenting friends or family members to act as buyers and ask a manager whether the resulting brief is useful.

3.2 Performance Metrics

Metric	Pilot target	How it will be measured
Valid event capture	At least 95%	Compare test WhatsApp events with the stored event ledger
Useful requirement summary	At least 80% accepted	Manager checks budget, BHK, area and timeline
Property grounding	100% catalog IDs	Compare every suggestion with the uploaded listings
Dashboard update delay	Under 5 seconds	Compare event and screen revision timestamps
STOP compliance	100%	Automated consent tests and manual pilot checks
Meeting brief usefulness	At least 80% accepted	Manager rates whether the brief supports a real visit

Table 3.1: Planned performance and safety measures for the pilot.

3.3 Analysis

The main success measure is **time to a useful manager action**: the time between a buyer's first valid WhatsApp message and a dashboard result that tells the manager what to ask, what properties to show or whether to arrange a visit. The project will not use message count alone as a success measure because a long conversation can still be unhelpful.

Figure 3.1 shows the rule for producing a meeting-ready lead. The AI may detect and record interest, but the final booking stays with the manager. This prevents a buyer from receiving a false promise about availability or timing.

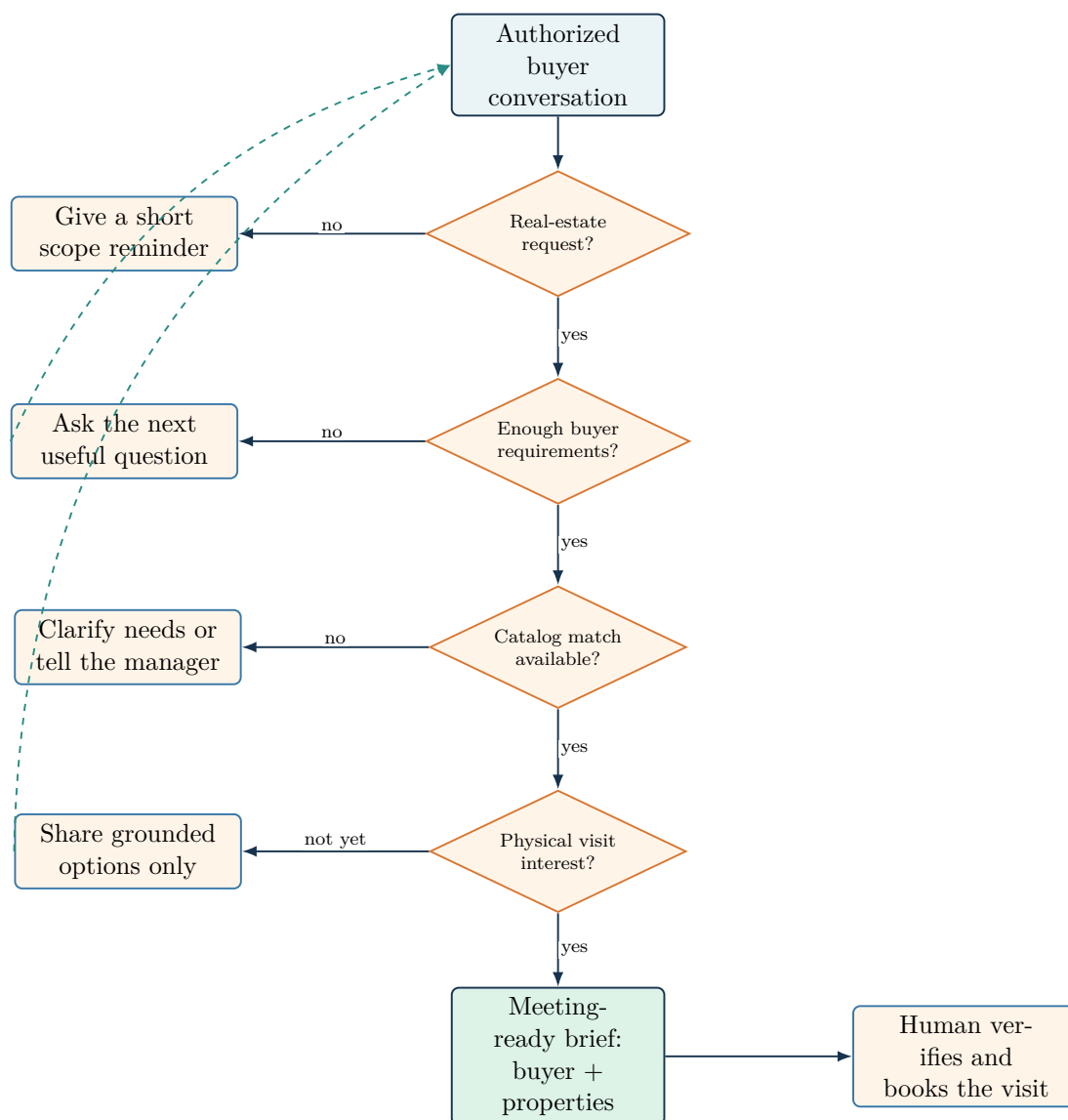


Figure 3.1: Decision flow for creating a physical-meeting-ready buyer brief.

The pilot results will be reviewed in three groups. First, technical results will show whether events arrive once and screens update quickly. Second, quality results will show whether buyer facts and suggested listings are correct. Third, safety results will show whether the system stays inside real estate, respects consent and leaves important commitments to a human.

Conclusion

4.1 Summary of Work

Estate Desk proposes a practical bridge between WhatsApp property conversations and manager action. It combines a restricted OpenClaw real-estate agent, an authorized CRM list, a property catalog, signed event handling and live web dashboards. The expected result is a short, useful buyer brief that tells the manager what the buyer wants, which uploaded properties fit and whether the buyer appears ready for a physical meeting.

4.2 Key Findings

- **The manager dashboard should support the conversation, not replace it.** Buyers can continue to use the familiar WhatsApp channel.
- **A useful lead needs both chat and listing context.** Chat alone cannot prove that a suitable property is actually in the manager's catalog.
- **Meeting readiness is an AI suggestion, not a confirmed booking.** A human must verify the property, time and buyer's final agreement.
- **Safety needs more than a prompt.** Contact authorization, consent, signed events and outgoing-message checks reduce avoidable misuse.

4.3 Future Work and Recommendations

1. connect a real CRM through its authorized API instead of CSV files;
2. add hosted WhatsApp connection management for multiple property teams;
3. add staff roles, lead assignment and calendar-based meeting requests;
4. connect verified live inventory and mark changed prices or availability;
5. evaluate extraction and matching on a larger, consented conversation set;
6. add stronger audit reports, retention controls and deletion tools; and
7. compare different approved models without changing the safety boundary.

4.4 Final Remarks

The project is suitable for a third-year software engineering team because it has a clear user problem, separate modules, measurable targets and visible safety limits. It also has a useful first version that can be tested with a small group before larger integrations are attempted. The main idea is simple: let the AI organize and qualify property interest, but let people control consent, property truth and the final physical meeting.

Bibliography

- [Clo26] Cloudflare. *Cloudflare D1 Documentation*. 2026. (Visited on 09/14/2026) (cit. on p. 8).
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